

SYSTEM ANALYSIS AND DESIGN

Unit-II: Feasibility Studies

Comprehensive Study Notes

1. Introduction to Feasibility Study

1.1 What is a Feasibility Study?

A **Feasibility Study** is a preliminary investigation to determine whether a proposed system is viable and worth pursuing. It assesses the technical, operational, economic, and behavioral aspects of the proposed system.

Definition (James O'Brien): "Feasibility study is an evaluation of a proposed project to determine whether it is technically, economically, and operationally viable."

Definition (Pressman): "Feasibility study is the process of identifying and evaluating the potential of a proposed system to meet the organization's needs."

1.2 Purpose of Feasibility Study

Purpose	Description
Determine Viability	Assess if the project is feasible
Identify Options	Explore alternative solutions
Evaluate Alternatives	Compare different approaches
Recommend Solution	Suggest the best approach
Justify Investment	Provide basis for decision-making
Resource Planning	Plan required resources

1.3 Importance of Feasibility Study

Importance	Description
Risk Reduction	Reduces project risks
Cost Savings	Avoids unnecessary investment
Resource Optimization	Optimal resource allocation
Decision Making	Informed decision-making
Project Success	Increases success probability
Time Savings	Saves time in the long run

2. Technical Feasibility

2.1 What is Technical Feasibility?

Technical Feasibility assesses whether the organization has the technical resources and capabilities to implement the proposed system.

2.2 Questions for Technical Feasibility

Question	Description
Hardware	Is the required hardware available?
Software	Is the required software available?
Skills	Do we have the required technical skills?
Capacity	Can existing systems handle the load?
Compatibility	Will the new system integrate with existing systems?
Technology	Is the technology mature and reliable?

2.3 Technical Feasibility Assessment

Factor	Assessment
Hardware Capacity	Sufficient capacity available
Software Availability	Required software available
Technical Skills	Skills available or can be acquired
System Integration	Can integrate with existing systems
Technology Maturity	Technology is stable and reliable

2.4 Solved Example: Technical Feasibility

Scenario: A company wants to implement a cloud-based CRM system.

Factor	Status
Hardware	Cloud-based, no new hardware needed
Software	Available from vendor
Skills	Need training, but skills are available in the market
Capacity	Scalable cloud solution
Compatibility	API integration possible
Technology	Mature cloud technology

Conclusion: Technically feasible.

3. Operational Feasibility

3.1 What is Operational Feasibility?

Operational Feasibility assesses whether the proposed system will work in the organization and whether users will accept it.

3.2 Questions for Operational Feasibility

Question	Description
User Acceptance	Will users accept and adopt the system?
Organizational Impact	How will the system affect the organization?
Management Support	Is management committed to the project?
Process Changes	How will it change business processes?
Training Needs	What training will users need?
Resistance	Is there likely to be resistance to change?

3.3 Operational Feasibility Assessment

Factor	Assessment
User Acceptance	High probability of acceptance
Organizational Impact	Positive impact expected
Management Support	Full management support
Process Changes	Acceptable level of change
Training	Training can be provided
Resistance	Minimal resistance expected

4. Behavioral Feasibility

4.1 What is Behavioral Feasibility?

Behavioral Feasibility assesses how the proposed system will affect people and whether they will accept and adapt to the changes.

4.2 Behavioral Feasibility Assessment

Factor	Assessment
User Attitudes	Positive attitude towards technology
Organizational Culture	Culture supports innovation
Change Management	Change management plan in place
Training	Training will be provided
Communication	Effective communication plan

5. Economic Feasibility

5.1 What is Economic Feasibility?

Economic Feasibility assesses whether the proposed system is financially viable and provides value for money.

5.2 Costs vs Benefits

Costs

Cost Type	Examples
Development Costs	Salaries, consultants, tools
Hardware Costs	Servers, workstations, networking
Software Costs	Licenses, development tools
Implementation Costs	Installation, data migration, training
Operational Costs	Maintenance, support, upgrades
Training Costs	User training, documentation

Benefits

Benefit Type	Examples
Tangible Benefits	Reduced staff, lower costs, increased sales
Intangible Benefits	Better customer service, improved image
Productivity Gains	Faster processing, less errors
Quality Improvements	Better quality products/services

5.3 Cost-Benefit Analysis

Cost-Benefit Analysis compares the costs and benefits of a proposed system.

5.4 Economic Analysis Techniques

Technique	Formula	Description
Payback Period	Initial Investment / Annual Cash Inflow	Time to recover investment
Net Present Value	Sum of PV of benefits - PV of costs	Value of future cash flows
Return on Investment	$(\text{Benefits} - \text{Costs}) / \text{Costs} \times 100$	Percentage return
Break-even Analysis	Fixed Costs / (Price - Variable Cost)	Point of no profit no loss

5.5 Solved Example: Cost-Benefit Analysis

Scenario: A company wants to implement a new inventory system.

Costs:

- Development: ₹3,00,000
- Hardware: ₹1,00,000
- Software: ₹50,000
- Training: ₹30,000

Benefits (Annual):

- Staff reduction: ₹1,50,000
- Reduced errors: ₹80,000
- Faster processing: ₹50,000

6. Questions

Short Answer Questions (2-5 Marks)

1. What is a feasibility study?
2. What are the types of feasibility?
3. What is technical feasibility?
4. What is operational feasibility?
5. What is economic feasibility?
6. What is behavioral feasibility?
7. What is cost-benefit analysis?
8. What is payback period?
9. What is ROI?
10. What is the importance of feasibility study?

Long Answer Questions (10 Marks)

1. Explain the different types of feasibility studies.
2. Explain technical feasibility in detail.
3. Explain operational feasibility in detail.
4. Explain economic feasibility and cost-benefit analysis.
5. Explain behavioral feasibility with examples.

Practice Questions

Q.No.	Question
Q1	Perform cost-benefit analysis for a proposed system
Q2	Assess technical feasibility for a new system
Q3	Assess operational feasibility for a new system
Q4	Calculate payback period and ROI
Q5	Conduct a feasibility study for a given scenario

Objective Questions (1 Mark)

1. Feasibility study is done in: (a) Planning phase (b) Design phase (c) Implementation phase (d) Testing phase
2. Technical feasibility assesses: (a) Technology (b) Finance (c) People (d) All
3. Operational feasibility assesses: (a) Technology (b) Finance (c) People and processes (d) All
4. Economic feasibility assesses: (a) Costs (b) Benefits (c) Both (d) None
5. Payback period formula is: (a) Investment / Annual Benefit (b) Annual Benefit / Investment (c) Cost / Benefit (d) None

Quick Revision

Feasibility Types Summary

Feasibility Study

-- Technical Feasibility	(Hardware, Software, Skills)
-- Operational Feasibility	(Users, Processes, Management)
-- Economic Feasibility	(Costs, Benefits, ROI)
-- Behavioral Feasibility	(Attitudes, Culture, Change)
-- Schedule Feasibility	(Timelines, Milestones)
-- Legal Feasibility	(Compliance, Regulations)

Key Formulas

Payback Period = Initial Investment / Annual Cash Inflow

ROI = (Total Benefits - Total Costs) / Total Costs x 100

Net Present Value = Sum of PV of Benefits - PV of Costs

Break-even Point = Fixed Costs / (Price - Variable Cost)

End of Unit-II Notes